Pooling and spatial information

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

Prepare the purpose, prerequisites and materials

> Prepare Pooling and spatial information: check the target, prerequisites, resources and adaptable teaching schedule.

Teach the learning cycle

Primary target: Compute max and **mean pooling** under an explicit window rule

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Retrieval answer guidance

Convolution as a local operation

Without opening the old lesson, explain one key idea from Convolution as a local operation. Give an example and a mistake that the idea helps you avoid.

Look for: Compute a local weighted sum at each valid image position

Reference example, not the only acceptable wording: Teacher explanation: Window rows are [2, 0] and [1, 3]. Products are 2, 0, 0 and 3, giving **output** 5.

Learned representations

Without opening the old lesson, explain one key idea from Learned representations. Give an example and a mistake that the idea helps you avoid.

Look for: Trace an **input** through a hidden representation and a decision rule

Reference example, not the only acceptable wording: Teacher explanation: The vectors are [5, 0], [0, 5] and [0, 0]. **Prediction**s are 1, 0 and 0. Equal width and height does not satisfy strictly greater.

Regularisation and early stopping

Without opening the old lesson, explain one key idea from Regularisation and early stopping. Give an example and a mistake that the idea helps you avoid.

Look for: Separate a weight penalty from **prediction** error

Reference example, not the only acceptable wording: Teacher explanation: MSE 0, penalty 2, total J 2, total gradient 2. Zero data gradient does not remove the penalty gradient.

For the compact alternative schedule, use two 45-minute sessions. Prepare a labelled 4 by 4 grid with four window outlines. State complete windows, stride 2 and no padding before any arithmetic.

Keep pooling separate from ReLU, and discuss loss of location through explicit collisions.

Reference for the floor-based pooling convention: https://docs.pytorch.org/docs/stable/generated/torch.nn.modules.pooling.MaxPool2d.html . No software library is required.

Outcomes

Compute max and **mean pooling** under an explicit window rule

Use collisions to explain lost **spatial information**

Test boundary and movement claims with revealing examples

Preparation

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

Prerequisite answer

Only the start at row 0 column 0 fits. The **output** shape is 1 by 1 under the complete-window rule.

End goal for the pictorial case

By the end, I can explain pooling and **spatial information** using a traced example and a stated limitation.

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Explain and model the mechanism

> Model Pooling and spatial information using the supplied case, labelled mechanism and reasoning. Make the scope of each conclusion explicit.

A fictional single-channel feature map has rows [1, 4, 2, 0], [3, 2, 5, 1], [0, 6, 1, 3] and [2, 4, 2, 7]. Use a 2 by 2 window, stride 2, no padding and only complete windows. Coordinates are zero-based.

The four starts are [0, 0], [0, 2], [2, 0] and [2, 2].

Max pooling keeps the largest value in each window. **Mean pooling** adds the four values and divides by 4. Neither operation learns weights in this lesson.

These are fixed summaries of the incoming feature map, unlike the trainable kernel parameters introduced last week.

The max **output** is [[4, 5], [6, 7]], and the mean **output** is [[2.5, 2], [3, 3.25]]. Both have shape 2 by 2. Sixteen values become four, but the four **output**s still preserve the coarse arrangement of the four regions.

This code assumes a nonempty rectangular numeric map at least 2 by 2. It uses only complete 2 by 2 windows, stride 2 and no padding.

It returns two separate summaries of the same map; it does not apply one pooling operation after the other.

For this complete-window rule the **output** shape is floor((H - 2) / 2) + 1 by floor((W - 2) / 2) + 1. A 3 by 3 map produces just one **output**; its last row and last column are not read.

Do not silently invent a partial-window rule or pad missing positions.

Term	Meaning
Max pooling	Replacing each window by its largest value
Mean pooling	Replacing each window by the arithmetic mean of its values
Spatial information	Information about where values occur and how positions relate
Pooling collision	Different inputs producing the same pooled values

A maximum does not identify its location

INPUT WINDOW ROWS	MAXIMUM	MEAN
[4 0] and [0 0]	4	1
[0 4] and [0 0]	4	1
[4 4] and [4 4]	4	4

The first two windows differ in the position of their nonzero value yet have the same maximum and mean. The third has the same maximum but a different mean. Each summary retains some distinctions and discards others.

Explain Pooling and spatial information · 1

Guided: both windows have maximum 6 and mean 1.5. Either scalar alone cannot distinguish their different nonzero positions.

Exit: the first two visual windows collide, so exact reconstruction is impossible from the pooled values alone. The 8 crossing from column 1 to column 2 changes the pooled map, so universal shift invariance does not follow.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

A maximum does not identify its location

INPUT WINDOW ROWS	MAXIMUM	MEAN
[4 0] and [0 0]	4	1
[0 4] and [0 0]	4	1
[4 4] and [4 4]	4	4

The first two windows differ in the position of their nonzero value yet have the same maximum and mean. The third has the same maximum but a different mean. Each summary retains some distinctions and discards others.

Compare windows [[0, 0], [6, 0]] and [[0, 0], [0, 6]]. Compute both summaries. Can a deterministic head receiving only either scalar distinguish bottom-left from bottom-right?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Pooling and spatial information

Pool a small map and show the detail it cannot retain.

The facts we will use

Supplied facts

Feature map rows [1, 4, 2, 0],[3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

Supplied facts

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-5/22

Two adjacent pooling windows

	left 0	left 1	right 0	right 1
row 0	1	4	2	0
row 1	3	2	5	1

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/22

Pooling and spatial information

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Left window

Values 1, 4, 3, 2: max 4, mean 2.5.

2. Right window

Values 2, 0, 5, 1: max 5, mean 2.

3. Compare summaries

Max map [[4, 5]], mean map [[2.5, 2]], both shape 1 x 2.

CHANGE ONE THING

Swap 1 and 4 within the left window.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Left window	Values 1, 4, 3, 2: max 4, mean 2.5.
2. Right window	Values 2, 0, 5, 1: max 5, mean 2.
3. Compare summaries	Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

What this example does not establish

Keep the stated limit

Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

Change one thing in the pictured case

Change one thing

Change: Swap 1 and 4 within the left window.

Predicted result

Prediction: Both pooled maps remain unchanged.

Why the result follows

Reason: Max and mean ignore the positions of values inside the same window.

Ask while showing the picture

Ask while showing the picture · 2

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

Demonstrate and guide practice

> Demonstrate Pooling and spatial information, reduce support deliberately and check individual reasoning before discussing answers.

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve valid window starts and mark the four nonoverlapping regions.
7 to 18 minutes	Calculate both summaries in the first region, then complete W1.
18 to 32 minutes	Trace the remaining regions and run or hand-trace the optional code.
32 to 45 minutes	Use W2 and the visual to distinguish lost location from retained regional arrangement.
Session 2 start to 8 minutes	Complete the guided collision and discuss why storing a maximum position is still insufficient for reconstruction.
8 to 20 minutes	Use W3 to expose the omitted border, then trace the movement counterexample in W5.
20 to 35 minutes	Collect W4 independently, requiring signed sums and both maps without applying ReLU.
35 to 45 minutes	Use the exit response to qualify claims, then administer the fresh retry.

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate columns for window values, sum, divisor, maximum and mean. Use numbered positions rather than colour alone.

Extension: enumerate all sixteen binary 2 by 2 windows, group them by maximum and mean, and identify collisions under each summary.

Reduce help deliberately

Completion practice

Completion task: Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

Reasoning to listen for

Teacher check: Teacher explanation: For example, [[4, 0], [0, 0]] and [[0, 4], [0, 0]] both give maximum 4 and mean 1.

A deterministic head receiving the same scalar must give the same **output**, so it cannot distinguish those positions.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual reasoning check

Use the worked case for Pooling and **spatial information**. Proposed explanation: "**Max pooling** turns negative numbers into zero". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected repair

Expected correction: That is a ReLU behaviour, not the maximum of an all-negative window.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Reasoned teacher answer

No. The original 4 was top-right in the left window, but after swapping it is top-left; the pooled value 4 is identical. Changed-case check: Both pooled maps remain unchanged.

Max and mean ignore the positions of values inside the same window.

Changed-case reasoning: Both pooled maps remain unchanged. Max and mean ignore the positions of values inside the same window.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

TEACHER GUIDE · 4 OF 9

Run the investigation and diagnose misconceptions

> Run the Pooling and spatial information investigation and use the first unsupported step to choose a teaching response.

Max pooling turns negative numbers into zero: That is a ReLU behaviour, not the maximum of an all-negative window.

Pooling removes all **spatial information**: A local pooled grid retains coarse regional arrangement while losing within-window detail.

Saving the maximum position restores the **input**: Other values remain unknown even when a unique maximum location is recorded.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Match the support to the first error

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

Assess and explain the answers

> Use the Pooling and spatial information questions, teacher keys and criteria after the relevant attempts. Previously discussed cases are practice; protect the separately selected independent task.

Guided and exit questions

Guided: both windows have maximum 6 and mean 1.5. Either scalar alone cannot distinguish their different nonzero positions.

Exit: the first two visual windows collide, so exact reconstruction is impossible from the pooled values alone. The 8 crossing from column 1 to column 2 changes the pooled map, so universal shift invariance does not follow.

W1 Calculate both summaries

Workbook question · W1

For the lesson window starting at [2, 2], list the four values, sum, maximum and mean. State the complete pooled **output** shape.

Teacher reasoning · W1

Teacher explanation: Values are 1, 3, 2 and 7. Sum is 13, maximum 7 and mean 3.25. Each pooled **output** has shape 2 by 2.

W2 Prove that location can be lost

Workbook question · W2

Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

Teacher reasoning · W2

Teacher explanation: For example, [[4, 0], [0, 0]] and [[0, 4], [0, 0]] both give maximum 4 and mean 1. A deterministic head receiving the same scalar must give the same **output**, so it cannot distinguish those positions.

W3 Audit an omitted border

Workbook question · W3

Apply the lesson complete-window rule to [[1, 2, 99], [3, 4, 99], [99, 99, 99]]. Give the **output** shape and both **output**s. Explain why neither **output** includes 99.

Teacher reasoning · W3

Teacher explanation: Shape is 1 by 1. Only [[1, 2], [3, 4]] is read, giving maximum 4 and mean 2.5. The final row and column lie outside the only complete stride -2 window.

W4 Pool a new map independently

Workbook question · W4

Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map. Explain why replacing a negative maximum by zero is an error here.

Teacher reasoning · W4

Teacher explanation: Window sums in row order are -10, 12, 16 and -20. Max map is [[-1, 6], [7, -2]]. Mean map is [[-2.5, 3], [4, -5]].

Zero is not in either all-negative window; pooling is not ReLU and must return the largest actual value.

W5 Check the movement boundary

Workbook question · W5

A 2 by 4 map is zero except for 8 at row 0 column 1. Move the 8 to column 2 and set its old

position to zero. Give the max and mean maps before and after. What claim does this refute?

Teacher reasoning · W5

Teacher explanation: Before: max [[8, 0]], mean [[2, 0]]. After: max [[0, 8]], mean [[0, 2]]. Crossing a pooling boundary changes the **output**, refuting guaranteed invariance to every one-pixel movement.

Independent assessment and fresh retry

Teacher key · Independent assessment and fresh retry

W4 requires sums -10, 12, 16, -20, max map [[-1, 6], [7, -2]] and mean map [[-2.5, 3], [4, -5]]. Fresh retry: rows [-6, -2, 1, 5] and [-4, -8, 3, 7]. **Output** shape is 1 by 2.

Sums are -20 and 16; max map is [[-2, 7]] and mean map is [[-5, 4]]. Require a signed maximum rather than a zero-clamped answer.

Assess independent transfer

Independent question · withhold the key

Selected case: Use rows [-4, -1, 2, 6], [-3, -2, 4, 0], [5, 1, -8, -2], [3, 7, -4, -6]. Under the lesson rule, give all four sums, the max map and the mean map.

Explain why replacing a negative maximum by zero is an error here.

Selected case answer

Teacher key · Selected case answer

Teacher explanation: Window sums in row order are -10, 12, 16 and -20. Max map is [[-1, 6], [7, -2]]. Mean map is [[-2.5, 3], [4, -5]].

> Zero is not in either all-negative window; pooling is not ReLU and must return the largest actual value.

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Teacher key · Selected case answer

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Feedback that the learner uses

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

Connect the project, support and next lesson

> Connect Pooling and spatial information to Image-pattern Network Investigation, adapt support and plan a later changed-case check.

Add a labelled pooling and **spatial information evidence** sheet to Image-pattern Network Investigation. Show how you can compute max and **mean pooling** under an explicit window rule.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate columns for window values, sum, divisor, maximum and mean. Use numbered positions rather than colour alone.

Extension: enumerate all sixteen binary 2 by 2 windows, group them by maximum and mean, and identify collisions under each summary.

Week 23 trains and evaluates a small image model with a declared baseline and data split.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

Plan the delayed return

Suggested return points in this level: ai-5/23, ai-5/25, ai-5/28. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Project evidence from this case

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Prepare, teach and assess

> Use this preparation summary alongside the detailed Pooling and spatial information lesson; it is a checklist rather than a replacement for the full sequence.

Teaching target

Teaching target: Compute max and **mean pooling** under an explicit window rule

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.

> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: Give two different 2 by 2 windows containing exactly one 4 and three zeros that have the same maximum and mean. Explain why either scalar alone cannot identify the position of the 4.

Supported practice answer

Reasoned practice answer: Teacher explanation: For example, [[4, 0], [0, 0]] and [[0, 4], [0, 0]] both give maximum 4 and mean 1.

A deterministic head receiving the same scalar must give the same **output**, so it cannot distinguish those positions.

Check every learner before discussion: Use the worked case for Pooling and **spatial information**. Proposed explanation: "**Max pooling** turns negative numbers into zero". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Repair the misconception

Misconception repair: That is a ReLU behaviour, not the maximum of an all-negative window.

Support without giving away the answer

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extend the reasoning

Extension: Swap 1 and 4 within the left window. Require a **prediction** and a reason before comparing. Expected result: Both pooled maps remain unchanged. Reason: Max and mean ignore the positions of values inside the same window.

Protect the independent attempt

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint

Project checkpoint: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

TEACHER GUIDE · 8 OF 9

Use feedback, labs and repair paths

> Use feedback, earlier review and portfolios purposefully. Apply the coding routine only when this lesson has an executable lab.

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

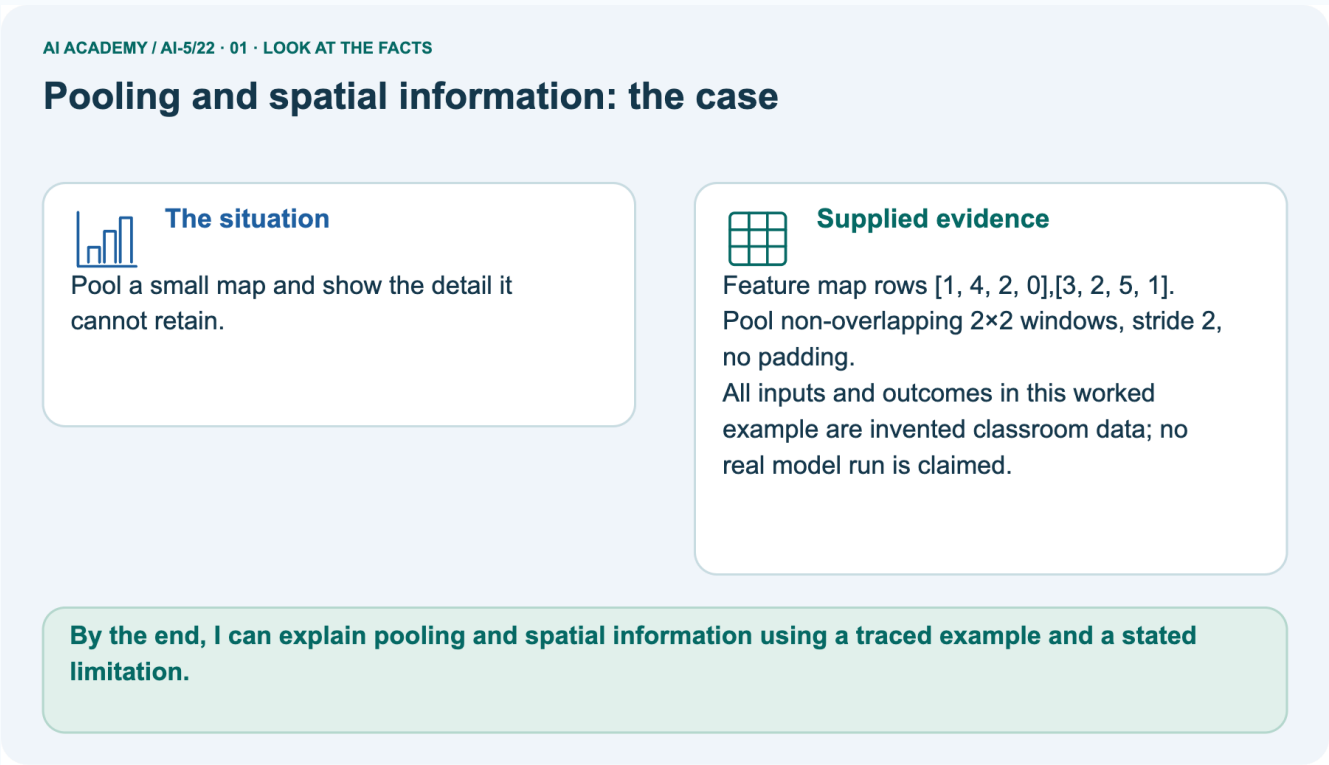
Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

Visual concept journey

> Show one Pooling and spatial information graphic at a time, request a prediction and discuss its trace before the independent assessment.

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Pooling and spatial information: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

Pool a small map and show the detail it cannot retain. Supplied facts: Feature map rows [1, 4, 2, 0], [3, 2, 5, 1]. Pool non-overlapping 2×2 windows, stride 2, no padding.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

Discuss while showing the picture

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Left window: Values 1, 4, 3, 2: max 4, mean 2.5. → Right window: Values 2, 0, 5, 1: max 5, mean 2. → Compare summaries: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2.

Point to each numbered stage and explain which recorded **input** it uses.

Discuss while showing the picture

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Two adjacent pooling windows



Follow the labelled picture with the matching explanation. Open the vector diagram

Two adjacent pooling windows. row 0: left 0 = 1, left 1 = 4, right 0 = 2, right 1 = 0. row 1: left 0 = 3, left 1 = 2, right 0 = 5, right 1 = 1.

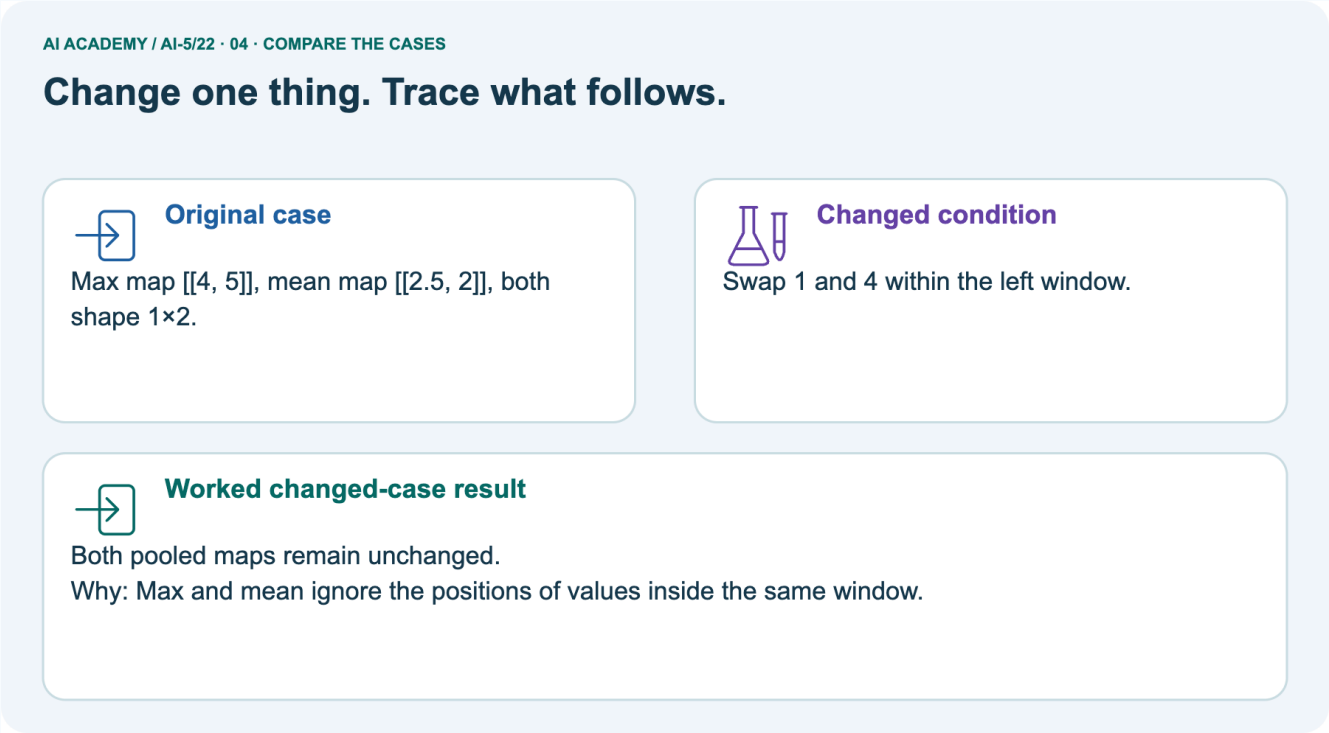
Read the labels and values. Explain the relationship the model shows.

Discuss while showing the picture

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Max map [[4, 5]], mean map [[2.5, 2]], both shape 1×2. Changed condition: Swap 1 and 4 within the left window. Expected result: Both pooled maps remain unchanged.

Reason: Max and mean ignore the positions of values inside the same window.

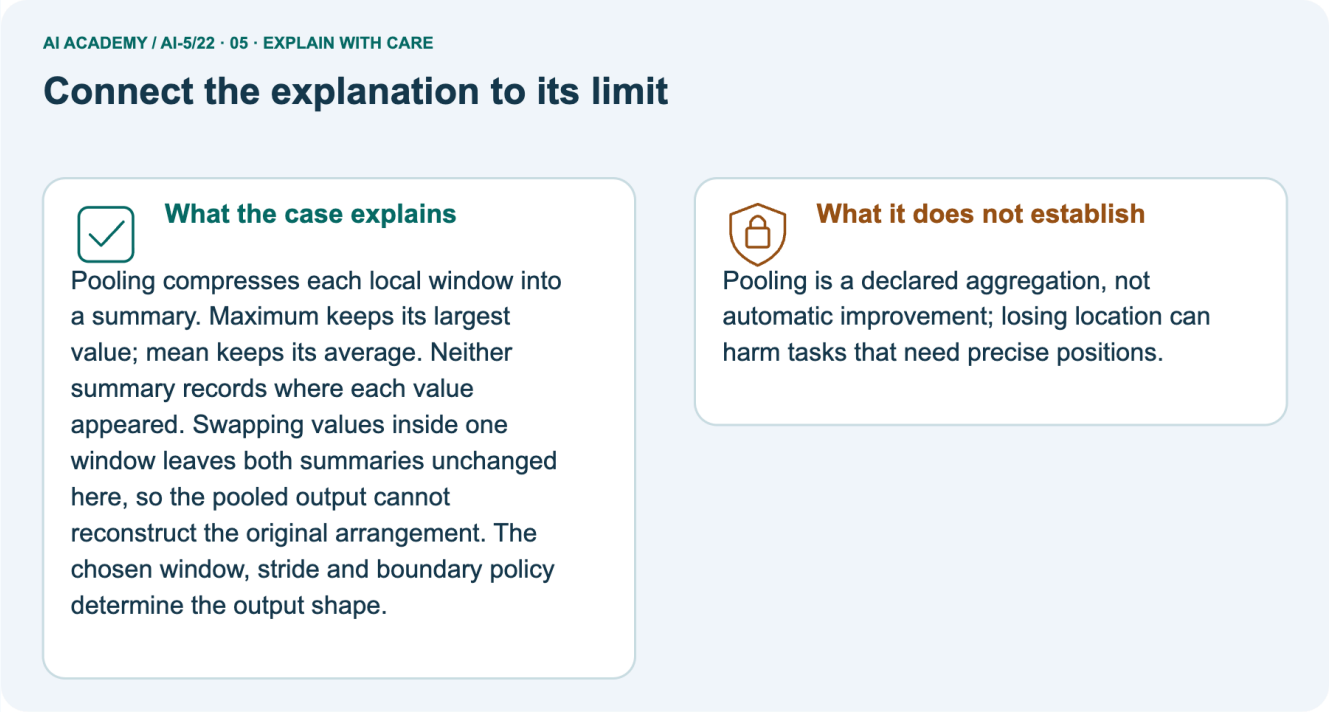
Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Discuss while showing the picture

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Max and mean ignore the positions of values inside the same window.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Pooling compresses each local window into a summary. Maximum keeps its largest value; mean keeps its average. Neither summary records where each value appeared.

Swapping values inside one window leaves both summaries unchanged here, so the pooled **output** cannot reconstruct the original arrangement. The chosen window, stride and boundary policy determine the **output** shape.

Limit: Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

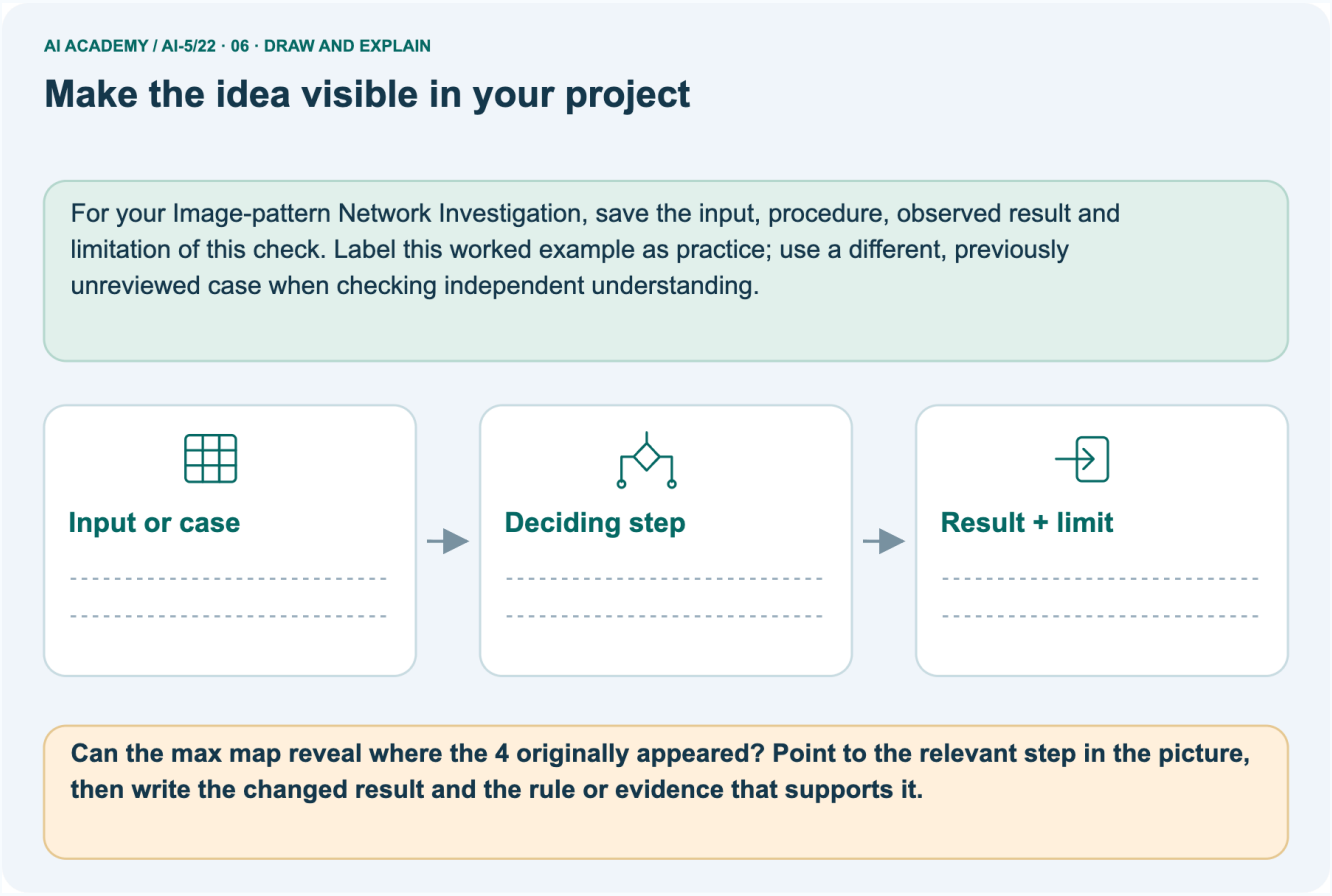
Explain the case, then point to a limit on the claim.

Discuss while showing the picture

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Pooling is a declared aggregation, not automatic improvement; losing location can harm tasks that need precise positions.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Can the max map reveal where the 4 originally appeared? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

> Draw your project trace and justify the deciding step using **evidence**.

Discuss while showing the picture

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. The original 4 was top-right in the left window, but after swapping it is top-left; the pooled value 4 is identical. Changed-case check: Both pooled maps remain unchanged.

Max and mean ignore the positions of values inside the same window.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.